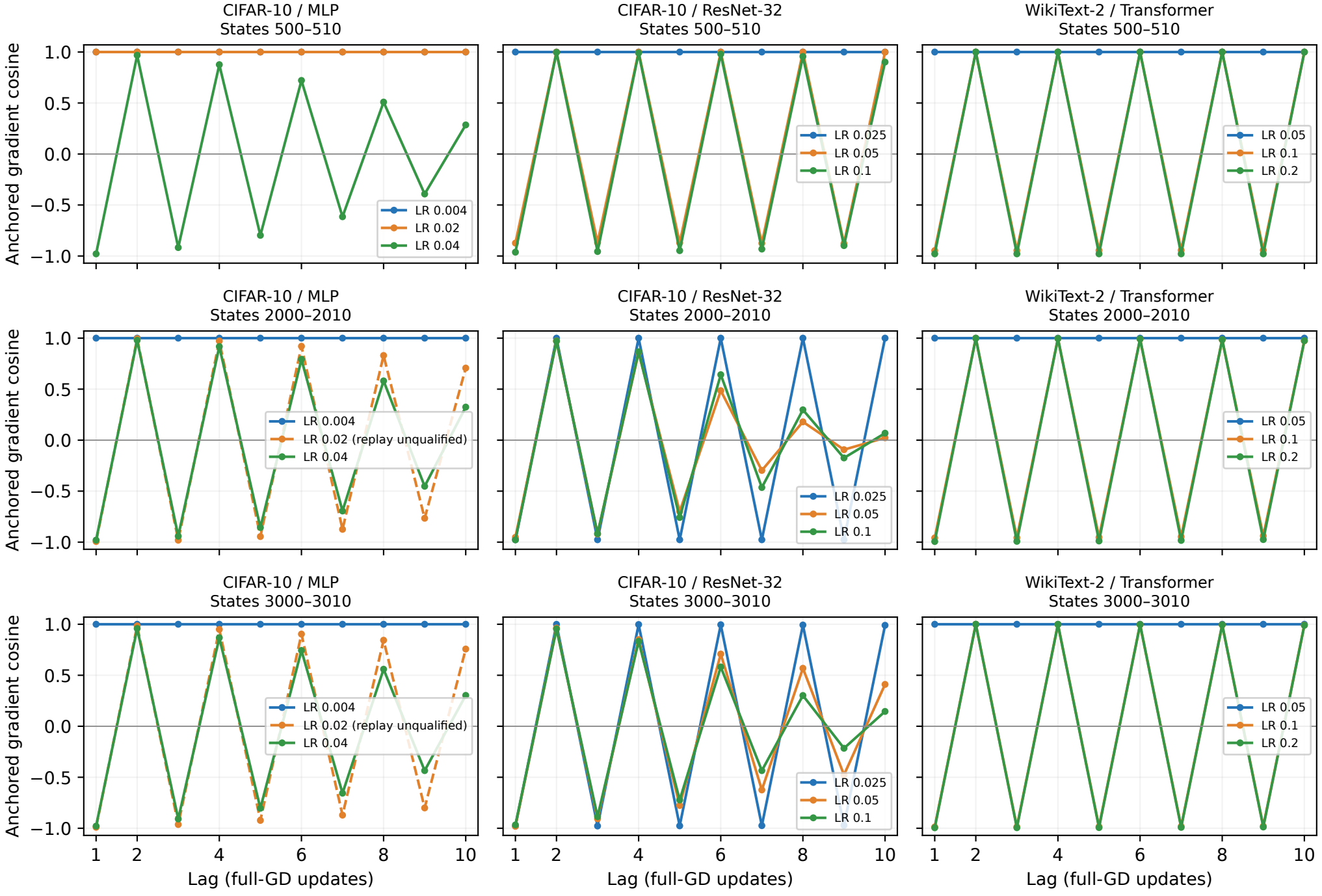

CIFAR-10 / MLP
States 500–510
CIFAR-10 / ResNet-32
States 500–510
WikiText-2 / Transformer
States 500–510
CIFAR-10 / MLP
States 2000–2010
CIFAR-10 / ResNet-32
States 2000–2010
WikiText-2 / Transformer
States 2000–2010
CIFAR-10 / MLP
States 3000–3010
CIFAR-10 / ResNet-32
States 3000–3010
WikiText-2 / Transformer
States 3000–3010
Anchored gradient cosine
1.0
0.5
0.0
−0.5
−1.0
LR 0.004
LR 0.02
LR 0.04
LR 0.02 (replay unqualified)
LR 0.025
LR 0.05
LR 0.1
LR 0.2
1
2
4
6
8
10
Lag (full-GD updates)